

Solutions: Motion hides in the timing

Instructor copy · Exercise session 2

PROBLEM 1

Three snapshots, one missing story

a) Reconstruct the motion. With $\vec{r}_0 = \vec{0}$, snapshots B and C give, in SI units,

$$\vec{v}_0 + \frac{1}{2}\vec{a} = (6, 2), \quad 3\vec{v}_0 + \frac{9}{2}\vec{a} = (0, 12).$$

Subtract three times the first equation from the second: $3\vec{a} = (-18, 6)$. Hence

$$\vec{v}_0 = (9, 1) \text{ m/s}, \quad \vec{a} = (-6, 2) \text{ m/s}^2.$$

For use below, with t in seconds,

$$x = 9t - 3t^2, \quad y = t + t^2, \quad v_x = 9 - 6t, \quad v_y = 1 + 2t.$$

b) A turn is not a stop. The rightmost position occurs where $v_x = 0$:

$$t = 1.5 \text{ s.}$$

Since $a_x < 0$, the robot moves right before this time and left after it. At the turn,

$$\vec{v} = (0, 4) \text{ m/s.}$$

It is still moving in the positive y direction, so it has not stopped.

c) Average velocity. Divide the displacement from A to C by the full elapsed time:

$$\vec{v} = \frac{(0, 12) - (0, 0)}{3} = (0, 4) \text{ m/s.}$$

Reasoning checkpoint. The snapshots are separated by 1 s and 2 s. They are not equally spaced in time.

PROBLEM 2

A throw through a moving window

a) Horizontal meeting. The screen is at $x_s = 8 + 2t$ and the ball at $x_b = 6t$, with SI units understood. Thus

$$6t = 8 + 2t \quad \Longrightarrow \quad \boxed{t_c = 2 \text{ s.}}$$

Both are then at $x = 12 \text{ m}$.

b) Clear the edges. The ball's height is $y = ut - 5t^2$, so at the crossing $y_c = 2u - 20$. The window requires

$$3 < 2u - 20 < 5 \quad \Longrightarrow \quad \boxed{11.5 \text{ m/s} < u < 12.5 \text{ m/s.}}$$

The inequalities are strict because touching an edge does not count. The ball returns to ground at $t_g = 2u/g$, between 2.3 s and 2.5 s, so every permitted launch is still airborne at the crossing.

c) Centre the throw. The centre is at $y = 4 \text{ m}$. Set $2u - 20 = 4$:

$$\boxed{u = 12 \text{ m/s.}}$$

At the window, the vertical velocity is

$$v_y = u - gt_c = 12 - 10 \cdot 2 = -8 \text{ m/s.}$$

The ball is **falling** as it passes through the centre.

Reasoning checkpoint. Match horizontal positions first. Then use that same time to check the height.

PROBLEM 3

The slowest possible interception

Optional challenge

a) Components and deadline. Use s uphill and n normally away from the ramp. The target and ball obey

$$\begin{aligned} s_T &= L - \frac{1}{2}g \sin \alpha t^2, & n_T &= 0, \\ s_B &= v_{0s}t - \frac{1}{2}g \sin \alpha t^2, & n_B &= v_{0n}t - \frac{1}{2}g \cos \alpha t^2. \end{aligned}$$

Equating their positions at the meeting gives

$$v_{0s} = \frac{L}{t}, \quad v_{0n} = \frac{g \cos \alpha}{2}t, \quad 0 < t < t_b = \sqrt{\frac{2L}{g \sin \alpha}}.$$

The upper endpoint is excluded: the target must not yet have reached O .

b) Minimum without the deadline. Write $c = g \cos \alpha / 2$. Then

$$v_0^2 = \frac{L^2}{t^2} + c^2 t^2 = \left(\frac{L}{t} - ct \right)^2 + 2Lc \geq gL \cos \alpha.$$

Equality requires $L/t = ct$, so

$$t_* = \sqrt{\frac{2L}{g \cos \alpha}}, \quad v_{0,\min} = \sqrt{gL \cos \alpha}.$$

The two launch components are equal: the direction is 45° above the ramp.

c) Restore the deadline. The optimum is possible precisely when

$$t_* < t_b \iff \tan \alpha < 1 \iff \alpha < 45^\circ.$$

For $\alpha \geq 45^\circ$, the speed decreases throughout the allowed interval: $L/t - ct$ is positive and decreases there. Its limiting value occurs as $t \rightarrow t_b^-$:

$$\begin{aligned} 0 < \alpha < 45^\circ : & \quad v_{0,\min} = \sqrt{gL \cos \alpha} \quad (\text{attained}), \\ 45^\circ \leq \alpha < 90^\circ : & \quad v_0 > \sqrt{\frac{gL}{2 \sin \alpha}} \quad (\text{bound never attained}). \end{aligned}$$

At 45° the optimum occurs exactly at the forbidden deadline. For steeper ramps it occurs too late. In both cases a successful launch can approach the bound as closely as desired, but there is no slowest one.

Flight check. For any allowed meeting time t , at an earlier time $0 < \tau < t$ the ball has $n_B(\tau) = \frac{1}{2}g \cos \alpha \tau(t - \tau) > 0$. It stays above the ramp until the meeting.